MATHEMATICS II FOR DATA SCIENCE
GA-7 SOLUTIONS MAY 2024

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1. Question

Consider the vector spaces V and functions $\langle .,. \rangle : V \times V \to \mathbb{R}$ defined as follows:

i) $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$.
ii) $V = M_{2\times 2}(\mathbb{R})$ and $\langle A, B \rangle = Tr(AB)$,
where $Tr(M)$ denotes the trace of a matrix M, i.e., the sum of the diagonal elements of M.
iii) $V = M_{2\times 1}(\mathbb{R})$ and $\langle A, B \rangle = Tr(AB^t)$,
where $Tr(X)$ denotes the trace of a matrix X, i.e., the sum of the diagonal elements of X. Y^t denotes the transpose of matrix Y.
iv) $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_2 + x_2y_1$.

- ☐ (i) is an inner product.
- ☐ (ii) is an inner product.
- ☐ (iii) is an inner product.
- ☐ (iv) is an inner product.

1.1. Answers:

- (i) is an inner product
- (iii) is an inner product

1.2. Solution

(i) $< (x_1, x_2), (y_1, y_2) > \;=\; x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$

- Check if $< v, v > \;>\; \forall v \setminus \{0\}; \;\; < v, v \geqslant= 0 \; if \; and \; only \; if \; v = 0$

Let $v = (x, y)$ such that $v \neq 0$

$$\begin{aligned} < v, v > \; &=< (x, y), (x, y) > \\ &= x^2 - xy - xy + 2y^2 \\ &= x^2 - 2xy + 2y^2 \\ &= \left(x^2 - 2xy + y^2\right) + y^2 \\ &= (x - y)^2 + y^2 \end{aligned}$$

Notice that $(x - y)^2 + y^2$ is a sum of two squared terms. Therefore, $< v, v >$ will be greater than 0 for all $v \neq 0$. We can also see that $< 0, 0 > \;=\; 0$

- Check if $< (x_1, x_2), (y_1, y_2) > \;=\; < (y_1, y_2), (x_1, x_2) >$

$$< (x_1, x_2), (y_1, y_2) > \;= x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$$

and,

$$< (y_1, y_2), (x_1, x_2) > \;= y_1x_1 - y_2x_1 - y_1x_2 + 2y_2x_2$$

we can see that both are equal.

- Check if $< u + v, w > \;=\; < u, w > + < v, w >$

Let $u = (a, b),\ v = (i, j),\ w = (x, y)$ where $a, b, i, j, x, y \in \mathbb{R}$

$< u + v, w >$	$< u, w > + < v, w >$
$=< (a, b) + (i, j), (x, y) >$ $=< (a + i, b + j), (x, y) >$ $= (ax + ai) - (bx + jx) - (ay + iy) + 2(by + jy)$	$=< (a, b), (x, y) > \;+\; < (i, j), (x, y) >$ $= ax - bx - ay + 2by + ix - jx - iy + 2jy$ $= (ax + ix) - (bx + jx) - (ay + iy_{+}2(by + jy)$

we can see that $LHS = RHS$.

- Check if $< cu, v > \;=\; c < u, v >$, where $c \in \mathbb{R}$ and $u = (a, b),\ v = (i, j)$ where $u, v, i, j) \in \mathbb{R}$

$< cu, v >$	$c < u, v >$
$=< c(a, b), (i, j) >$ $=< (ca, cb), (i, j) >$ $= cai - cbi - caj + 2cbj$	$= c < (a, b), (i, j) >$ $= c(ai - bi - aj + 2bj)$ $= cai - cbi - caj + 2bj$

we can see that $LHS = RHS$.

Therefore, $< (x_1, x_2), (y_1, y_2) > \;=\; x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$ is an inner product. Hence, **option 1** is correct.

(ii) $V = M_{2\times2}(R)$ and $< A, B > \;=\; Tr(AB)$

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ then $A^2 = \begin{bmatrix} -1 & 4 \\ -2 & -1 \end{bmatrix}$. We can see that $Tr\left(A^2\right) < 0$. This would mean that $< A, A > \;> 0\ \forall\ A \setminus \{0\}$ is not satisfied. $< A, B > \;=\; Tr(AB)$ is not an inner product. Hence, **option 2** is incorrect.

(iii) $V = M_{2\times1}(R)$ and $< A, B > \;=\; Tr\left(AB^t\right)$

Let $A = \begin{bmatrix} a & b \end{bmatrix},\ B = \begin{bmatrix} i & j \end{bmatrix},\ C = \begin{bmatrix} x & y \end{bmatrix},\ where\ A, B, C \in M_{2\times2}(\mathbb{R})$

- Check if $< A, A > \ > 0 \ for\ all\ A \setminus \{0\}$. $< A, A > \ = 0$ if and only if $A = 0$.

$$\begin{aligned} < A, A > &= A \times A^t \\ &= \begin{bmatrix} a & b \end{bmatrix} \times \begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} a^2 + b^2 \end{bmatrix} \end{aligned}$$

$Tr\left(AA^t\right) = a^2 + b^2$. We can see that $Tr\left(AA^t\right)$ is a sum of squared terms, therefore will be greater than 0 for all $A \setminus \{0\}$. Notice that $< A, A > \ = \ 0$ when $A = \begin{bmatrix} 0 & 0 \end{bmatrix}$.

- Check if $< A, B > \ = \ < B, A >$

$AB^t = [ai + bj]$ and $BA^t = [ia + bj]$.
$T\left(AB^t\right) = ai + bj = Tr\left(BA^t\right)$. Therefore, $< A, B > \ = \ < B, A >$

- Check if $< A + B, C > \ = \ < A, C > + < B, C >$

$< A + B, C >$	$< A, C > + < B, C >$
$=< \begin{bmatrix} a & b \end{bmatrix} + \begin{bmatrix} i & j \end{bmatrix}, \begin{bmatrix} x & y \end{bmatrix} >$ $=< \begin{bmatrix} a+i & b+j \end{bmatrix}, \begin{bmatrix} x & y \end{bmatrix} >$ $= \begin{bmatrix} a+i & b+j \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$ $= ax + ix + by + jy$	$=< \begin{bmatrix} a & b \end{bmatrix}, \begin{bmatrix} x & y \end{bmatrix} > \ + \ < \begin{bmatrix} i & j \end{bmatrix}, \begin{bmatrix} x & y \end{bmatrix} >$ $= \begin{bmatrix} a & b \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} i & j \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$ $= ax + ix + by + jy$

We can see that $LHS = RHS$

- Check if $< cA, B > \ = \ c < A, B >$, where $c \in \mathbb{R}$

$< cA, B >$	$< A, C > + < B, C >$
$=< c\begin{bmatrix} a & b \end{bmatrix}, \begin{bmatrix} i & j \end{bmatrix} >$ $=< \begin{bmatrix} ca & cb \end{bmatrix}, \begin{bmatrix} i & j \end{bmatrix} >$ $= \begin{bmatrix} ca & cb \end{bmatrix} \times \begin{bmatrix} i \\ j \end{bmatrix}$ $= cai + cbj$	$= c < \begin{bmatrix} a & b \end{bmatrix}, \begin{bmatrix} i & j \end{bmatrix} >$ $= c\begin{bmatrix} a & b \end{bmatrix} \times \begin{bmatrix} i \\ j \end{bmatrix}$ $= c(ai + bj)$ $= cai + cbj$

We can see that $LHS = RHS$.
Therefore, $< A, B > \ = \ Tr\left(AB^t\right)$ is an inner product. Hence, **option 3** is correct.

(iv) $V = R^2$ and $< (x_1, x_2), (y_1, y_2) > \ = \ x_1y_2 + x_2y_1$

Let $v = (1, 2)$ and $u = (3, 4)$ and $w = (5, 6)$.

$$
\begin{aligned}
< v+u, w > &=< (1,2)+(3,4),(5,6) > \\
&=< (3,6),(5,6) > \\
&= (3\times 6)+(6\times 5) \\
&= 48
\end{aligned}
$$

$$
\begin{aligned}
< v,w > + < u,w > &=< (1,2),(5,6) > \ + \ < (3,4),(5,6) > \\
&= (1\times 6)+(2\times 5)+(3\times 6)+(4\times 5) \\
&= 6+10+18+20 \\
&= 54
\end{aligned}
$$

We can see that $< v+u, w > \neq < v,w > + < u,w >$ in this case.
Therefore, $< (x_1,x_2),(y_1,y_2) > \ = \ x_1y_2 + x_2y_1$ is not an inner product. Hence, **option 4** is incorrect.

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2. Question

Consider two linear transformations T and S from $\mathbb{R}^2 \to \mathbb{R}^2$ defined as $T(x,y) = (2x+y, x+y)$ and $S(x,y) = (x+cy, x+2y)$. Let A and B be matrix representations of linear transformations T and S with respect to the standard bases of $\mathbb{R}^2$ respectively.
Consider the following statements:

P: If $c = 1$, then A and B are similar matrices.
Q: If $c = 2$, then A and B are similar matrices.
R: If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$.
S: If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.
T: If $c = 1$, then there are infinitely many P satisfying the equation $P^{-1}AP = B$.

Which of the following options are true?

- ☐ P is true but Q is false.
- ☐ Both P and Q are true.
- ☐ Both R and S are true.
- ☐ R is false but S is true.
- ☐ T is true.

2.1. Answers :

- P is true but Q is false.
- R is false but S is true.
- T is true.

2.2. Solution

Let us find matrix A

$$\begin{aligned} T(1,0) &= (2,1) \\ &= 2(1,0) + 1(0,1) \end{aligned}$$

$$\begin{aligned} T(0,1) &= (1,1) \\ &= 1(1,0) + 1(0,1) \end{aligned}$$

$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

Let us find matrix B

$$\begin{aligned} T(1,0) &= (1,1) \\ &= 1(1,0) + 1(0,1) \end{aligned}$$

$$\begin{aligned} T(0,1) &= (c,2) \\ &= c(1,0) + 2(0,1) \end{aligned}$$

$B = \begin{bmatrix} 1 & c \\ 1 & 2 \end{bmatrix}$

The reduced row echelon form of A is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

If $c = 2$, then $B = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$. In this case, the reduced row echelon form of B would be $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$.

From this we can see that $rank(A) = 2$ and $rank(B) = 1$. If two matrices don't have the same rank, then they cannot be similar. Hence, **statement Q** is incorrect.

If $c = 1$, then $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$. In this case, the reduced row echelon form of B us $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

From this we can see that $rank(B) = 2 = rank(A)$. However, we cannot conclude that A and B are similar. For A and B to be similar, we need an invertible matrix P such that the following holds true:

$$B = P^{-1}AP$$

If there is at least one invertible matrix P which satisfies $B = P^{-1}AP$, then we can say that A and B are invertible.

Let $P = \begin{bmatrix} a & b \\ k & d \end{bmatrix}$ where $a, b, k, d \in \mathbb{R}$ and $det(A) \neq 0$.

$$\begin{aligned} B &= AP \\ PB &= AP \end{aligned}$$

$$\begin{bmatrix} a & b \\ k & d \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ k & d \end{bmatrix}$$

$$\begin{bmatrix} a+b & a+2b \\ k+d & k+2d \end{bmatrix} = \begin{bmatrix} 2a+k & 2b+d \\ a+k & b+d \end{bmatrix} \quad (1)$$

From (1), we get the following equations.

$$\begin{aligned} a+b &= 2a+k \\ a+k-b &= 0 \qquad (2) \end{aligned}$$

$$\begin{aligned} a+2b &= 2b+d \\ d &= a \qquad (3) \end{aligned}$$

$$\begin{aligned} k+d &= a+k \\ d &= a \qquad (4) \end{aligned}$$

$$\begin{aligned} k+2d &= b+d \\ k+d-b &= 0 \qquad (5) \end{aligned}$$

Substitute $d = a$ in (5), we get $k + a - b = 0$. This is the same as (2).
We get $k = b - a$

Matrix $P = \begin{bmatrix} a & b \\ b-a & a \end{bmatrix}$.
Notice that $det(P) = a^2 + ab - b^2$.
$det(P) \neq 0$
$a^2 + ab - b^2 \neq 0$
$a^2 + ab \neq b^2$

We can have infinitely many solutions for a and b such that $a^2 + ab \neq b^2$. Therefore, **statement P** and **statement T** are correct.

Can $P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$? Equate $\begin{bmatrix} a & b \\ b-a & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. We can see that $a = 1,\ b = 0$. But $b - a \neq 1$. Hence, **statement R** is incorrect.

Can $P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$? Equate $\begin{bmatrix} a & b \\ b-a & a \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$. We can see that $a = 1,\ b = 2$. We can also see that $b - a = 1$. Hence, **statement S** is correct.

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3. Question

Let $\langle .,. \rangle$ denote the standard inner product or $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$. Which one of the following options is (are) true for the vector $\gamma \in \mathbb{R}^2$, such that $\langle \alpha, \gamma \rangle = 4$ and $\langle \beta, \gamma \rangle = 8$, where $(3, 1)$ and $\beta = (6, 2)$.

- ☐ No such γ exists.
- ☐ There are infinitely many such vectors which satisfy the properties of γ.
- ☐ γ is unique in $\mathbb{R}^2$.
- ☐ Any vector in the set $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$ satisfies the properties of γ.
- ☐ $(1, 1)$ is the only vector which satisfies the properties of γ.

3.1. Answers :

- There are infinitely many such vectors which satisfy the properties of γ.
- Any vector in the set $\{(t, 4 - 3t) | t \in \mathbb{R}\}$ satisfies the properties of γ.

3.2. Solution

Let $\gamma = (t, u)$.

$$\begin{aligned} < \alpha, \gamma > &= 4 \\ < (3, 1), (t, u) > &= 4 \\ 3t + u &= 4 \quad (1) \end{aligned}$$

$$\begin{aligned} < \beta, \gamma > &= 8 \\ < (6, 2), (t, u) > &= 8 \\ 6t + 2u &= 8 \\ 3t + u &= 4 \quad (2) \end{aligned}$$

(1) and (2) are the same. There are infinite solutions for $3t + u$. Therefore, there are infinitely many such vectors which satisfy the properties of γ. Hence, **option 1** and **option 3** are incorrect and **opiton 2** is correct.

The solution for $3p + u = 4$ is the following set : –
$\{(t, u) | t, u \in \mathbb{R} \; and \; 3t + u = 4\} \implies \{(t, 4 - 3t) | t \in \mathbb{R}\}$. Hence, **option 4** is correct.

Notice that $(1, 1)$ is not of the form $(t, 4 - 3t)$. Hence, **option 5** is incorrect.

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4. Question

Let $\langle .,. \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ and let $v \in \mathbb{R}^2$. Consider a linear transformation $T_v : \mathbb{R}^2 \to \mathbb{R}$ defined as:

$$T_v(u) =< u, v >$$

where $v \in \mathbb{R}^2$. Which of the following options is (are) true for T_v?

- ☐ T_v is one-one for all $v \neq 0 \in \mathbb{R}^2$.
- ☐ T_v is onto for all $v \neq 0 \in \mathbb{R}^2$.
- ☐ T_v is onto for all $v \in \mathbb{R}^2$.
- ☐ T_v is not one-one for every $v \in \mathbb{R}^2$.
- ☐ There exists a $v \in \mathbb{R}^2$ such that T_v is not onto.

4.1. Answers :

- T_v is onto for all $v \neq 0$.
- T_v is not one-one for every $v \in \mathbb{R}^2$.
- There exists a $v \in \mathbb{R}^2$ such that T_v is not onto.

4.2. Solution

Let $\{(1,0),(0,1)\}$ be a basis for $\mathbb{R}^2$ and let $\{1\}$ be a basis for $\mathbb{R}$. Let $v = (p,q)$, where $v \in \mathbb{R}^2$ and $v \neq 0$

$$\begin{aligned} T_v(1,0) &=< (1,0),(p,q) > \\ &= (1 \times p) + (0 \times q) \\ &= p(1) \end{aligned}$$

$$\begin{aligned} T_v(0,1) &=< (0,1),(p,q) > \\ &= (0 \times p) + (1 \times q) \\ &= q(1) \end{aligned}$$

from the above, we can see that the matrix representation of T_v with respect to $\{(1,0),(0,1)\}$ form $\mathbb{R}^2$ and $\{1\}$ for $\mathbb{R}$ is : –

$$\begin{bmatrix} p & q \end{bmatrix}$$

The reduced row echelon form for $\begin{bmatrix} p & q \end{bmatrix}$ when $p \neq 0$ is $\begin{bmatrix} 1 & q/p \end{bmatrix}$.

If $p = 0$ then $q \neq 0$ as $v \neq 0$. In this case, the reduced row echelon form form $\begin{bmatrix} p & q \end{bmatrix}$ is $\begin{bmatrix} 0 & 1 \end{bmatrix}$.
In both cases, $rank(T_v) = 1$ and $nullity(T_v) = 1$.

A linear transformation is one-to-one if and only if its nullity is 0. Therefore, T_v is not one-to one.
A linear transformation is onto if and only if its rank is equal to the dimension of its codomain. $dim(\mathbb{R}) = 1 = rank(T_v)$. Therefore, T_v is onto,

If $v = 0$, $T_v(u) = 0$ for all $u \in \mathbb{R}^2$. This is because, the given inner product is standard inner product on $\mathbb{R}^2$. For such an inner product, $< (x, y), (p, q) > \; = \; xp + yq$. Therefore, $< u, 0 > \; = \; 0$ for all $u \in \mathbb{R}^2$. In such a case, the range of T_v will be $\{0\}$. In this case $rank(T_v) = 0 \neq dim(\mathbb{R})$, where $\mathbb{R}$ is the codomain of T_v. But we know that a linear transformation is onto if and only if its rank is equal to the dimension of its codomain.

Hence, **option 2**, **option 4** and **option 5** are correct. **Option 1** and **option 3** are incorrect.

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5. Question

Let L and L' be affine subspaces of $\mathbb{R}^3$, where $L = (0, 1, 1) + U$ and $L' = (0, 1, 0) + U'$, for some vector subspaces U and U' of $\mathbb{R}^3$. Let a basis for U be given by $\{(1, 1, 0), (1, 0, 1)\}$ and a basis for U' be given by $\{(1, 0, 0)\}$. Suppose there is a linear transformation $T : U \to U'$ such that $(1, 0, 1) \in ket(T)$ and $T(1, 1, 0) = (1, 0, 0)$. An affine mapping $f : L \to L'$ is obtained by $f((0, 1, 1) + u) = (0, 1, 0) + T(u)$, for all $u \in U$. Which of the following options are true?

- ☐ $L = \{(x, y + 1, x - y + 1) \mid x, y \in \mathbb{R}\}$.
- ☐ $L' = \{(x, y + 1, 0) \mid x, y \in \mathbb{R}\}$.
- ☐ $L = \{(x - y, y + 1, x - y + 1) \mid x, y \in \mathbb{R}\}$.
- ☐ $L = \{(x, x + 1, y + 1) \mid x, y \in \mathbb{R}\}$.
- ☐ $f(x, y + 1, x - y + 1) = (y, 1, 0)$
- ☐ $f(x - y, y + 1, x - y + 1) = (x, y + 1, 0)$
- ☐ $f(x, x + 1, y + 1) = (y, 1, 0)$
- ☐ $f(x, y + 1, x - y + 1) = (0, 1, y)$

5.1. Answer :

- $L = \{(x, y + 1, x - y + 1) | x, y \in \mathbb{R}\}$
- $f(x, y + 1, x - y + 1) = (y, 1, 0)$

5.2. Solution

Every vector in U is of the form (x, y, z), where $x, y, z \in \mathbb{R}$ and it can be expressed as a linear combination of $\{(1, 1, 0), (1, 0, 1)\}$. This set is a basis for U . Let $a, b \in \mathbb{R}$.

$$(x, y, z) = a(1, 1, 0) + b(1, 0, 1)$$
$$(x, y, z) = (a + b, a, b)$$

We can see that $y = a$ and $z = b$. From this we can write x as : –

$$x = a + b \rightarrow x = y + z \rightarrow x - y = z$$

$$\begin{aligned}(x, y, z) &= y(1, 1, 0) + z(1, 0, 1) \\ (x, y, z) &= y(1, 1, 0) + (x - y) \cdot (1, 0, 1) \quad (1)\end{aligned}$$

U can be written as : –

$$\begin{aligned}U &= \{(x, y, z) | x, y, z \in \mathbb{R} \; and \; x = y + z\} \\ U &= \{(x, y, z) | x, y, z \in \mathbb{R} \; and \; z = x - y\} \\ U &= \{(x, y, x - y) | x, y \in \mathbb{R}\}\end{aligned}$$

Now, we can write L as : –

$$\begin{aligned}L &= (0, 1, 1) + U \\ L &= (0, 1, 1) + \{(x, y, x + y) | x, y \in \mathbb{R}\} \\ L &= \{(x, y + 1, x - y + 1) | x, y \in \mathbb{R}\}\end{aligned}$$

Hence, **option 1** is correct and **option 3** and **option 4** are incorrect.

Similarly, every vector in U' is in the form (x, y, z), where $x, y, z \in \mathbb{R}$ and can be expressed as a linear combination of $\{(1, 0, 0)\}$.This is a basis for U'. Let $p \in \mathbb{R}$.

$$\begin{aligned}(x, y, z) &= p(1, 0, 0) \\ (x, y, z) &= (p, 0, 0)\end{aligned}$$

We can see that $x = a, \; y = 0, \; z = 0$.
U' can be written as : –

$$\begin{aligned}U &= \{(x, y, z) | x = p, \; y = 0, \; z = 0, \; p \in \mathbb{R}\} \\ U &= \{(x, 0, 0) | x \in \mathbb{R}\}\end{aligned}$$

Now, L' can be written as : –

$$\begin{aligned}L' &= (0, 1, 0) + \{(x, 0, 0) | x \in \mathbb{R}\} \\ L' &= \{(x, 1, 0\} | x \in \mathbb{R}\}\end{aligned}$$

Hence, **option 2** is incorrect.

It is given that $T(1, 1, 0) = (1, 0, 0)$ and $T(1, 0, 1) = (0, 0, 0)$

From (1) we get : –

$$\begin{aligned}(x, y, z) &= y(1, 1, 0) + (x - y) \cdot (1, 0, 1) \\ T(x, y, z) &= T(y(1, 1, 0) + (x - y) \cdot (1, 0, 1)) \\ T(x, y, z) &= y \cdot T(1, 1, 0) + (x - y) \cdot T(1, 0, 1) \\ T(x, y, z) &= y \cdot (1, 0, 0) + (x - y) \cdot (0, 0, 0)\end{aligned}$$

$$T(x, y, z) = (y, 0, 0)$$

$f((0, 1, 1) + u) = (0, 1, 0) + T(u)$

Let $u = (x, y, z)$ and $u \in U$. We know that $z = x - y$ for all $(x, y, z) \in U$. Therefore, $u = (x, y, x - y)$

$$\begin{aligned} f((0, 1, 1) + u) &= (0, 1, 0) + T(u) \\ f((0, 1, 1) + (x, y, x - y)) &= (0, 1, 0) + T(x, y, z) \\ f(x, y + 1, x - y + 1) &= (0, 1, 0) + (y, 0, 0) \\ f(x, y + 1, x - y + 1) &= (y, 1, 0) \end{aligned}$$

Hence, **option 6** is correct. **Option 5**, **option 7** and **option 8** are incorrect.

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6. Question

Let θ be the angle between the vectors $u = (4, 7, 3)$ and $v = (1, 2, -6)$, then what will be the value of $cos(\theta)$?

6.1. Answer : 0

6.2. Solution

Angle(θ) between two vectors (v, w) is given as follows : –

$$\theta = cos^{-1}\left(\frac{v \cdot u}{||v|| \times ||u||}\right)$$

Where $\cdot$ denotes the dot product and $||v||$ denotes the length of v.

$$\begin{aligned} \theta &= cos^{-1}\left(\frac{v \cdot u}{||v|| \times ||u||}\right) \\ \theta &= cos^{-1}\left(\frac{(4, 7, 3) \cdot (1, 2, -6)}{\sqrt{(4, 7, 3) \cdot (4, 7, 3)} \times \sqrt{(1, 2, -6) \cdot (1, 2, -6)}}\right) \\ \theta &= cos^{-1}\left(\frac{4 + (7 \times 2) + (3 \times - 6)}{\sqrt{(4 \times 4) + (7 \times 7) + (3 \times 3)} \times \sqrt{1 + (2 \times 2) + (3 \times - 6)}}\right) \\ \theta &= cos^{-1}\left(\frac{0}{\sqrt{(4 \times 4) + (7 \times 7) + (3 \times 3)} \times \sqrt{1 + (2 \times 2) + (3 \times - 6)}}\right) \\ \theta &= cos^{-1}(0) \\ \theta &= 90^\circ \\ cos(\theta) &= cos(90^\circ) \\ cos(\theta) &= 0 \end{aligned}$$

7. Question

Consider a basis

$$\{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)$$

of $\mathbb{R}^3$ with usual inner product. Suppose $v = \left(x, y, \dfrac{3x+y}{5}\right) \in V$ is written as $v = c_1v_1 + c_2v_2 + c_3v_3$, such that $c_1 + c_2 = 4$. What will be the value of c_3?

7.1. Answer : 2

7.2. Solution

$$\begin{aligned} v &= c_1v_1 + c_2v_2 + c_3v_3 \\ \left(x, y, \frac{3x+y}{5}\right) &= c_1(1, 2, 0) + c_2(2, -1, 0) + c_3(0, 0, 2) \\ \left(x, y, \frac{3x+y}{5}\right) &= (c_1 + 2c_2, 2c_1 - c_2 + c_3) \end{aligned}$$

Substitute $c_1 + c_2 = 4 \implies c_1 = 4 - c_2$,

$$\left(x, y, \frac{3x+y}{5}\right) = (4 + c_2, 8 - 3c_2 + c_3)$$

We get the following equations.

$$\begin{aligned} x &= 4 + c_2 \qquad (1) \\ y &= 8 - 3c_2 \qquad (2) \end{aligned}$$

$$\begin{aligned} \frac{3x+y}{5} &= 2c_3 \\ 3x + y &= 10c_3 \qquad (3) \end{aligned}$$

Substitute (1) and (2) in (3)

$$\begin{aligned} 3(4 + c_2) + (8 - 3c_2) &= 10c_3 \\ 12 + 3c_2 + 8 - 4c_3 &= 10c_3 \\ 20 &= 10c_3 \\ c_3 &= 2 \end{aligned}$$

8. Question

Let $V = \mathbb{R}^2$ be a vector space. Consider two inner products $\langle ., . \rangle_1$ and $\langle ., . \rangle_2$ on V defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1x_2 - x_1y_2 - x_2y_1 + 4y_1y_2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3x_1x_2 + 2y_1y_2.$$

If $\langle (a, b), (5, 5) \rangle_1 = 75$ and $\langle (a, b), (5, 5) \rangle_2 = 125$, then find the value $a + 2b$.

8.1. Answer : 15

8.2. Solution

Given $< (x_1, y_1), (x_2, y_2 >_1 = x_1y_1 - x_1y_2 - x_2y_1 + 4x_2y_2$,

$$\begin{aligned} < (a, b), (5, 5) >_1 &= 75 \\ 5a - 5a - 5b + 20b &= 75 \\ 15b &= 75 \\ b &= 5 \end{aligned}$$

Given $< (x_1, y_1), (x_2, y_2) >_2 = 3x_1x_2 + 2y_1y_2$,

$$\begin{aligned} < (a, b), (5, 5) >_2 &= 125 \\ 15a + 10b &= 125 \\ 15a &= (125 - 10b) \end{aligned}$$

Substitute $b = 15$ in $15a = (125 / 10b)$

$$\begin{aligned} 15a &= (125 - (10 \times 5)) \\ 15a &= 75 \\ a &= 5 \end{aligned}$$

Find $a + 2b$

$$\begin{aligned} a + 2b &= 5 + 2(5) \\ &= 15 \end{aligned}$$

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9. Question

Let $V = \mathbb{R}^3$ be the inner product space with usual inner product. If θ is the angle between $(14, 2, 10)$ and (a, b, c) where $14a + 2b + 10c = 0$ and $\left(a^2 + b^2 + c^2\right) \neq 0$, then find the value of $cos(\theta)$.

9.1. Answer : 0

9.2. Solution

Given $14a + 2b + 10c = 0$. Notice that the dot product between $(14, 2, 10)$ and (a, b, c) is $14a + 2b + 10c$. If two vectors are perpendicular, then the dot product of those two vectors will be 0. Therefore, $(14, 2, 10)$ and (a, b, c) are perpendicular.

For any two vectors v and w, the following holds true : –

$$cos^{-1}\left(\frac{v \cdot w}{||v|| \times ||w||}\right) = the\ angle\ between\ v\ and\ w$$

Here, $\cdot$ denotes dot product and $||v||$ is the length of v.
Given that θ is the angle between $(14, 2, 10)$ and (a, b, c), and $a^2 + b^2 + c^2 \neq 0$, we can write the following.

$$\begin{aligned}
cos^{-1}\left(\frac{(14, 2, 10) \cdot (a, b, c)}{||(14, 2, 10)|| \times ||(a, b, c||}\right) &= \theta \\
\frac{14a + 2b + 10c}{\sqrt{14^2 + 2^2 + 10^2} \times \sqrt{a^2 + b^2 + c^2}} &= cos(\theta) \\
\frac{0}{\sqrt{14^2 + 2^2 + 10^2} \times \sqrt{a^2 + b^2 + c^2}} &= cos(\theta) \\
0 &= cos(\theta)
\end{aligned}$$

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10. Question

Consider a vector $v = (x - 20, 2, 1) \in \mathbb{R}^3$. Find the value of x so that the length of the vector v is minimum.

10.1. Answer : 20

10.2. Solution

Length of vector v is given by $||v||$, where

$$||v|| = \sqrt{v \cdot v}$$

where $\cdot$ is the dot product.

$$\begin{aligned}
||v|| &= \sqrt{(x - 20)^2 + 2^2 + 1^2} \\
||v|| &= \sqrt{x^2 + 400 - 40x + 4 + 1} \\
||v|| &= \sqrt{x^2 - 40x + 409}
\end{aligned}$$

Notice that $\sqrt{x^2 - 40x + 409}$ is the square root of a quadratic expression. $x^2 - 40x + 409$ is of the form $ax^2 + bx + c$.
Here, $a = 1$. If $a > 0$, then the quadratic expression will have a global minima at its vertex. $||v||$ will be minimum at the vertex of $x^2 - 40x + 409$.
The derivative at the vertex with respect to x will be 0.
Therefore, $||v||$ will be minimum when $\frac{d}{dx}\left(x^2 - 40x + 409\right) = 0$.

$$\frac{d}{dx}\left(x^2 - 40x + 409\right) = 0$$

$$
\begin{aligned}
2x - 40 &= 0 \\
x &= 20
\end{aligned}
$$

Therefore, $||v||$ will be minimum when $x = 20$.